Random Effects Models

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1 Overview

Objectives and overview

- This material is based Chapter 11–12 on the textbook "A First Course in Design and Analysis of Experiments" (Oehlert, 2000).
- Chapter outcomes include:
  - Students will be able to distinguish between a random effect and a fixed effect.
  - Students will be able to identify when random effects are appropriate.
  - Students will know how to implement random effects and mixed effects models.

Introduction

- So far, the treatment effects that we have been working with can be classified as *fixed-effects*: we wish to estimate the effect of specific treatments.
- What if the treatments themselves are a random selection from a larger population?
- **Example:** A farmer has 100 pieces of equipment, and would like to know how the equipment used may impact crop-yield. Rather than testing all 100 pieces of equipment, they *randomly select* g pieces of equipment, and performs a CRD with these equipment.

Random Effects
A random effect is appropriate to model treatments that are a *random sample* from a population of potential treatments.
Random effects are useful if the focus of inference is on the *population*, not the individual treatment effects.

2 Models for Random Effects

Example: Carton factory

- We will build intuition of Random Effects models and designs using an example.

Carboard company
Suppose a company makes cardboard cartons for storing goods. They have 50 machines to make cartons, and want to understand the variation in quality for finished products.
They choose 10 machines at random from the 50, and make 40 cartons on each machine. The resulting cartons are tested for strength.

- This is a completely randomized design (CRD), with $g = 10$ treatments (the different machines), and $n_i = 40$, for a total of $N = 400$ EUs.
- The model we have been using for this type of data is:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \quad \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2).$$

- We are still going to use a very similar model. However, we are not as interested in the exact α_i for the 10 randomly chosen machines. Instead, we want to make inference on the larger population.
- We could proceed as usual, but that only tells us about the 10 specific machines that were randomly chosen, and we would get an entirely different set of machines if we repeated the experiment.
- Thus, we want to introduce a *random effect*: let the effect α_i represent a random draw from a population of possible effects:

$$y_{ij} = \mu + \alpha_i + \epsilon_{ij}, \qquad \alpha_i \overset{iid}{\sim} N(0, \sigma^2_\alpha), \quad \epsilon_{ij} \overset{iid}{\sim} N(0, \sigma^2).$$

- We no longer assume $\sum_i \alpha_i = 0$, rather $E[\alpha_i] = 0$.
- We have introduced a new parameter, σ^2_α, which describes the spread of the effects.
- *We assume that the two random components are independent.* That is, α_i and ϵ_{ij} are independent.
- If they are independent, the variance of the observed data is increased:

$$\text{Var}(y_{ij}) = \text{Var}(\mu + \alpha_i + \epsilon_{ij}) = \sigma^2_\alpha + \sigma^2.$$

- The parameters σ^2_α and σ^2 are called *variance components*.
- If observations belong to the same treatment group, then they now have non-zero correlation:

$$\text{Cor}(y_{ij}, y_{ik}) = \begin{cases} 1 & j = k \\ \sigma^2_\alpha/(\sigma^2_\alpha + \sigma^2) & j \neq k \end{cases}$$

- We call this within treatment group correlation the *intraclass* correlation.

Key summary

- In a random effects model, the parameters are μ, σ^2, and σ^2_α.
- We are not interested in making inferences about the specific α_i and ϵ_{ij}, which are treated as random variables, not parameters.
- Inference wise, we are now interested in confidence intervals and hypothesis tests regarding μ and σ^2.

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References

Oehlert GW (2000). *A First Course in Design and Analysis of Experiments*. W. H. Freeman. ISBN 0-7167-3510-5. 1